

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2 FT	(a)			Microwaves / Infra-red / X-rays Accept IR, don't accept micro or X 3 correct = 2 marks, 1 or 2 correct = 1 mark	2			2		
	(b)			[In a transverse wave] the vibrations / oscillations (1) are at <u>90° / right angles / perpendicular to</u> [the direction of] energy [transfer] / <u>the wave</u> (1) don't accept to and fro, back and fore, side to side unless 90° clearly explained	2			2		
	(c)			2 nd and 3 rd boxes ticked (1 mark for each) i.e. All em waves travel at the same speed in a vacuum (1) All em waves are a form of radiation (1) Lose 1 mark for each extra tick		2		2		
	(d)			Gamma rays have a high[er] frequency or energy or short[er] wavelength (1) So are [more] ionising (1) N.B. I disagree must be included to be awarded 2 marks Alternative: Radio waves or they have a low[er] frequency or energy or long[er] wavelength (1) So don't ionise (1) N.B. I disagree must be included to be awarded 2 marks Don't accept reference to power			2	2		
				Question 2 total	4	2	2	8	0	0